

HODLR for Gaussian-Process linear algebra in OpenTURNS

Speed and accuracy benchmark report

hodlr branch — consolidated from `python/test/results/*.csv` and `PLAN.md`

September 2026

Abstract

This report consolidates the speed and accuracy measurements of the hierarchical off-diagonal low-rank (HODLR) implementation of this branch, compared against the dense LAPACK Cholesky backend, the classic hmat-oss `HMAT` backend, and the hmat-oss `HMAT_HODLR` schur-style HODLR backend exposed by PR #3272. The central accuracy gate (relative solve error of the HODLR approximation versus the dense reference) **passes**: 4.75×10^{-3} for $n = 4001$, 1D Matern($\text{corr}=0.7$), well below the 10^{-2} threshold. A direct speed comparison between the two HODLR implementations is available from the combined benchmark CSV and is highlighted in Section 4.

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1 Scope and protocol

The benchmark script `python/test/benchmark_hmat_hodlr.py` sweeps:

- dimensions 1D–4D on $[-1, 1]^d$ (regular IntervalMesher grids);
- covariance models Matern($\nu = 2.5$) and Exponential, correlation lengths 0.1 and 0.01;
- matrix sizes $n \approx 1000, 2000, 5000, 10000, 20000$;
- four backends: LAPACK (dense), `HMAT` (hmat-oss, standard admissibility), `HMAT_HODLR` (hmat-oss HODLR via PR #3272), and `HODLR` (self-contained implementation of this branch).

Timings recorded per configuration: first GP realization (`t_first`, includes factorization), second realization (`t_second`, factorization reused), Karhunen-Loève P1 (`t_kl`, SPECTRA eigensolver on HODLR/HMAT-compressed storage), `GaussianProcessFitter` (`t_fitter`), and `GaussianProcessRegression` (`t_regression`).

Two installs were measured with the *same* script, sweep, machine and session (2026-09-13):

- **hodlr-branch**: this branch (`build/install`); all four backends.
- **pr3272**: the hmat-oss PR #3272 build (`openturns-pr-3272/build/install`); backends LAPACK, HMAT, HMAT_HODLR. The self-contained HODLR backend does not exist there (all its rows are N/A), so the cross-build backend comparison is *branch-HODLR vs pr3272-HMAT_HODLR* (Section 4).

Caveat: the branch build measured on 09-13 predates the pivot-floor accuracy fix (finalized 2026-09-16, see Section 2), so several large- n self-contained HODLR entries are dominated by the (now fixed) near-SPD heal/retry cascade.

2 Accuracy results

2.1 The accuracy gate

The critical gate compares the HODLR factorization against the dense LAPACK Cholesky reference on a 1D Matern($\text{corr}=0.7$) mesh, solving $C_H x = b$ for the same right-hand side in both representations (`harness /tmp/hodlr_acc.time.py`; criterion $\text{relerr} = \|x_H - x_d\|/\|x_d\| < 10^{-2}$):

measure	value	verdict
n	4001	—
leaf size	250	—
factorize time	0.192 s (idle re-run)	—
relative solve error	4.747×10^{-3}	✓ $< 10^{-2}$
regularization shift	1.7×10^{-2}	healthy heal band

The gate **passes** and is load-independent (dense-vs-HODLR). The largest n runs were also sane after the fix: $n = 20002$ `factorize`=1.809 s (idle gate), `solve`=8.1 ms, $y_0 = 2.966$, shift 1.7×10^{-2} .

2.2 The accuracy bug fixed in this session (near-SPD corrupt leaves)

Root cause found by bisection (2026-09-16): the leaf Cholesky (`dpotrf`) could return `info==0` while still holding a pivot at roundoff scale ($\text{minpivot}^2 \sim 1.7 \times 10^{-16}$). The “successful” factor was numerically garbage: corrected leaf cross terms $W = L_{00}^{-1}V \sim 10^8\text{--}10^{16}$, $K = W^T W \sim 10^{16}\text{--}10^{32}$, cascading into UK and corrupting the whole factorisation. Symptom:

	pre-fix (corrupt)	post-fix
1D $\text{corr}=0.7$ $n = 20001$ factorize	✗ H-err $\sim 5.3\text{e}34$ (5.13/5.3 in records) 21.7 s (polluted) / 544 s (poisoned)	✓ relerr $4.75\text{e} - 3$ ✓ ≈ 1.8 s (idle)

Fix: the heal loop now also rejects tiny-pivot `dpotrf` and escalates `healLambda` until $\text{minpivot}^2 \geq \max(\text{PivotFloorFactor} \cdot \text{kernelDiagMax}, 10^{-12})$. New ResourceMap key `HODLRMatrix-PivotFloorFactor` (`openturns.conf.in`: $1.0\text{e} - 4$, later tuned to $1.0\text{e} - 2$ so the heal lands in the healthy pivot band $\sim 4.3\text{e} - 3 \dots 3.4\text{e} - 2$); `healLambdaInit` declared.

2.3 Open accuracy items

- **2D 141×141 corr=0.1** ($n = 19881$): factorize 3.3 s, relative residual 7.9×10^{-17} — the corrupt-lambda overlap was *not* reproduced on this path; the case stays healthy but is parked as a separate documented near-SPD/singular-overlap issue (leaves heal at the nugget floor), not touched by this fix.
- **No persisted Litvinenko suite:** `benchmark_gp_full.py --accuracy (rel_norm2, gap2, KL, logdet)` exists but its output was never saved; only the PLAN.md gate numbers above survived (the harness `/tmp/hodlr_acc_time.py` is no longer in `/tmp`).
- **No accuracy comparison between the two HODLR implementations:** no dense-vs-HMAT-HODLR (PR #3272) relative-error measurement exists in any report or CSV. Only the speed-side comparison is available (Section 4).

3 Speed results (hodlr-branch build, all backends)

Full per-dimension sweeps of the branch build. Columns: first/second realization, KL, fitter, regression; – = not measured (LAPACK skipped above its affordability limits, or exception).

3.1 1D

Matern model

Table 1: 1D matern corr=0.1 — hodlr-branch build

method	n	1st	2nd	KL	fit	regr
LAPACK	1001	20.8399ms	180.7213 μ s	2.2520s	4.8509ms	212.4310 μ s
	2001	87.8720ms	404.5963 μ s	6.0350s	2.6839ms	187.1586 μ s
	5001	–	–	–	2.3808ms	190.0196 μ s
	10001	–	–	–	2.6436ms	180.4829 μ s
	20001	–	–	–	5.1110ms	202.1790 μ s
HMAT	1001	50.6859ms	251.0548 μ s	37.1575ms	6.7847ms	60.7967 μ s
	2001	38.8081ms	542.4023 μ s	71.1985ms	6.6464ms	44.3459 μ s
	5001	70.5607ms	995.1591 μ s	146.8034ms	6.6078ms	41.7233 μ s
	10001	152.6802ms	2.2230ms	316.4821ms	10.2813ms	66.9956 μ s
	20001	358.4502ms	4.4467ms	688.4832ms	10.0005ms	62.4657 μ s
HMAT_HODLR	1001	21.2550ms	190.9733 μ s	29.0554ms	6.6702ms	66.5188 μ s
	2001	34.8620ms	369.3104 μ s	64.8904ms	7.2701ms	52.6905 μ s
	5001	64.5852ms	844.7170 μ s	144.9242ms	6.9914ms	80.3471 μ s
	10001	173.9833ms	2.0039ms	318.6467ms	9.8956ms	57.6973 μ s
	20001	338.2170ms	4.3659ms	672.0607ms	10.1166ms	60.0815 μ s
HODLR	1001	10.5488ms	211.4773 μ s	26.6383ms	4.1211ms	92.7448 μ s
	2001	11.9703ms	328.0640 μ s	52.2609ms	4.0100ms	79.3934 μ s
	5001	44.0936ms	879.5261 μ s	134.0339ms	4.2150ms	64.3730 μ s
	10001	282.3546ms	2.0306ms	3.5674s	19.5074ms	106.5731 μ s
	20001	26.0115s	3.9177ms	26.3912s	8.3027ms	107.0499 μ s

Table 2: 1D matern corr=0.01 — hodlr-branch build

method	n	1st	2nd	KL	fit	regr
LAPACK	1001	94.9111ms	195.5032 μ s	698.7784ms	8.0078ms	212.1925 μ s
	2001	162.3385ms	1.9441ms	12.0483s	8.9025ms	195.7417 μ s
	5001	—	—	—	2.6808ms	185.9665 μ s
	10001	—	—	—	5.7607ms	233.8886 μ s
	20001	—	—	—	12.7215ms	265.3599 μ s
HMAT	1001	22.9859ms	292.3012 μ s	166.7645ms	6.2976ms	61.2736 μ s
	2001	39.2177ms	494.2417 μ s	350.4634ms	6.7003ms	68.9030 μ s
	5001	66.5741ms	1.2007ms	819.7272ms	6.4483ms	58.6510 μ s
	10001	161.1905ms	2.6064ms	1.6508s	9.9761ms	83.6849 μ s
	20001	389.9126ms	4.6277ms	2.9577s	8.4648ms	59.6046 μ s
HMAT_HODLR	1001	14.8900ms	182.1518 μ s	155.9377ms	6.1176ms	54.1210 μ s
	2001	33.6757ms	354.5284 μ s	311.3160ms	6.2504ms	73.6713 μ s
	5001	62.1877ms	1.0386ms	797.5798ms	6.2811ms	46.0148 μ s
	10001	212.7934ms	2.7173ms	1.5951s	10.2944ms	75.8171 μ s
	20001	381.6979ms	4.6675ms	3.0069s	8.6930ms	65.3267 μ s
HODLR	1001	6.4704ms	146.1506 μ s	147.0857ms	4.0705ms	86.3075 μ s
	2001	12.4528ms	318.5272 μ s	290.0660ms	4.2429ms	77.7245 μ s
	5001	30.5357ms	972.9862 μ s	797.7509ms	3.8755ms	64.8499 μ s
	10001	113.8282ms	2.8882ms	1.5992s	7.4894ms	113.4872 μ s
	20001	195.0102ms	4.6923ms	3.3123s	11.4009ms	110.8646 μ s

Exponential model

Table 3: 1D exponential corr=0.1 — hodlr-branch build

method	n	1st	2nd	KL	fit	regr
LAPACK	1001	31.6205ms	2.0897ms	1.6808s	3.8838ms	267.9825 μ s
	2001	28.0147ms	1.3599ms	8.5663s	2.6281ms	178.0987 μ s
	5001	—	—	—	2.4390ms	187.1586 μ s
	10001	—	—	—	2.0814ms	189.3044 μ s
	20001	—	—	—	4.1625ms	221.4909 μ s
HMAT	1001	20.4070ms	345.2301 μ s	58.3856ms	9.8894ms	77.2476 μ s
	2001	27.9570ms	506.8779 μ s	90.4386ms	8.0268ms	57.4589 μ s
	5001	37.6792ms	934.3624 μ s	188.9453ms	6.7728ms	54.8363 μ s
	10001	87.6999ms	2.3823ms	498.9746ms	18.2433ms	118.2556 μ s
	20001	264.8706ms	5.0249ms	771.1811ms	10.3314ms	65.0883 μ s
HMAT_HODLR	1001	17.1552ms	238.8954 μ s	49.2518ms	8.3878ms	59.8431 μ s
	2001	27.0355ms	435.5907 μ s	78.2433ms	6.8350ms	61.9888 μ s
	5001	37.2522ms	876.9035 μ s	169.8935ms	6.3367ms	47.4453 μ s
	10001	155.4320ms	2.6503ms	439.0540ms	11.6963ms	75.8171 μ s
	20001	266.9787ms	5.0952ms	783.7491ms	10.4489ms	68.9030 μ s
HODLR	1001	9.0213ms	184.2976 μ s	43.6378ms	4.9298ms	92.5064 μ s
	2001	13.4776ms	344.2764 μ s	71.5711ms	4.4901ms	90.5991 μ s
	5001	29.0082ms	782.0129 μ s	176.3484ms	4.0741ms	87.2612 μ s
	10001	111.4185ms	2.6596ms	421.5465ms	11.7509ms	103.4737 μ s
	20001	209.5237ms	4.7941ms	836.4038ms	8.3401ms	107.2884 μ s

Table 4: 1D exponential corr=0.01 — hodlr-branch build

method	n	1st	2nd	KL	fit	regr
LAPACK	1001	19.4016ms	189.3044 μ s	2.2193s	6.9318ms	313.9973 μ s
	2001	25.4765ms	382.6618 μ s	11.8110s	2.6715ms	190.0196 μ s
	5001	—	—	—	2.2252ms	207.9010 μ s
	10001	—	—	—	2.3255ms	189.5428 μ s
	20001	—	—	—	8.9931ms	255.8231 μ s
HMAT	1001	21.2166ms	326.6335 μ s	177.6540ms	5.5375ms	53.4058 μ s
	2001	31.6479ms	489.9502 μ s	260.6030ms	4.8647ms	54.1210 μ s
	5001	37.1075ms	1.0068ms	599.2453ms	4.5173ms	52.2137 μ s
	10001	88.0830ms	2.3689ms	1.1403s	7.1437ms	63.1809 μ s
	20001	280.4322ms	4.9198ms	2.3823s	7.2782ms	66.0419 μ s
HMAT_HODLR	1001	12.0006ms	241.9949 μ s	129.6363ms	4.7851ms	47.2069 μ s
	2001	22.4037ms	453.2337 μ s	232.1174ms	4.5640ms	45.7764 μ s
	5001	35.0933ms	716.9247 μ s	568.4183ms	4.3893ms	66.0419 μ s
	10001	139.0102ms	2.5570ms	1.1544s	7.1235ms	60.3199 μ s
	20001	264.8673ms	4.8883ms	2.3808s	7.3056ms	65.5651 μ s
HODLR	1001	7.0734ms	162.3631 μ s	116.0200ms	6.7821ms	136.1370 μ s
	2001	12.8524ms	327.1103 μ s	216.7389ms	3.8013ms	69.1414 μ s
	5001	31.1267ms	872.6120 μ s	584.2237ms	6.1977ms	77.9629 μ s
	10001	111.7826ms	2.5234ms	1.1775s	13.1605ms	149.2500 μ s
	20001	210.1967ms	4.5869ms	2.4704s	17.2839ms	128.0308 μ s

3.2 2D

Matern model

Table 5: 2D matern corr=0.1 — hodlr-branch build

method	n	1st	2nd	KL	fit	regr
LAPACK	1089	17.2575ms	264.4062 μ s	2.6315s	3.9127ms	315.6662 μ s
	2116	88.5930ms	461.3400 μ s	—	6.6204ms	248.1937 μ s
	5184	—	—	—	2.6374ms	169.9924 μ s
	10201	—	—	—	2.3804ms	209.5699 μ s
	20164	—	—	—	5.7139ms	294.9238 μ s
HMAT	1089	29.3431ms	309.4673 μ s	144.1455ms	6.4464ms	70.8103 μ s
	2116	112.0870ms	704.7653 μ s	242.5411ms	5.8405ms	52.4521 μ s
	5184	238.8456ms	1.2181ms	498.8730ms	4.4732ms	44.5843 μ s
	10201	467.8061ms	3.0799ms	1.1292s	6.1867ms	69.6182 μ s
	20164	1.0717s	6.0699ms	2.4024s	8.1284ms	74.6250 μ s
HMAT_HODLR	1089	21.2538ms	235.5576 μ s	106.7433ms	5.4507ms	44.8227 μ s
	2116	62.0856ms	418.1862 μ s	186.2407ms	5.0595ms	47.9221 μ s
	5184	194.2804ms	1.0691ms	481.5748ms	4.2722ms	51.2600 μ s
	10201	525.3742ms	3.0634ms	1.1469s	6.4907ms	80.5855 μ s
	20164	1.0338s	6.2084ms	2.3460s	8.1329ms	72.4792 μ s
HODLR	1089	18.3501ms	257.7305 μ s	112.0167ms	6.3672ms	97.2748 μ s

continued

method	n	1st	2nd	KL	fit	regr
	2116	55.7771ms	590.8012 μ s	232.1019ms	5.1210ms	89.4070 μ s
	5184	273.1204ms	2.0568ms	843.8439ms	5.4948ms	86.0691 μ s
	10201	905.1511ms	4.7724ms	2.1841s	17.9672ms	188.8275 μ s
	20164	2.6493s	10.7076ms	5.5858s	14.1592ms	185.9665 μ s

Table 6: 2D matern corr=0.01 — hodlr-branch build

method	n	1st	2nd	KL	fit	regr
LAPACK	1089	152.0307ms	278.4729 μ s	3.5572s	12.3968ms	450.1343 μ s
	2116	718.4174ms	2.7299ms	—	24.7061ms	452.5185 μ s
	5184	—	—	—	4.2725ms	209.3315 μ s
	10201	—	—	—	4.0793ms	168.8004 μ s
	20164	—	—	—	8.4114ms	279.4266 μ s
HMAT	1089	28.5017ms	380.5161 μ s	369.7240ms	4.0371ms	42.9153 μ s
	2116	84.9829ms	846.1475 μ s	726.2216ms	7.8142ms	50.0679 μ s
	5184	157.4779ms	1.0905ms	2.3554s	9.6250ms	60.0815 μ s
	10201	388.5632ms	3.0026ms	5.7970s	10.5145ms	65.0883 μ s
	20164	1.8627s	6.8672ms	15.7141s	12.9809ms	83.9233 μ s
HMAT_HODLR	1089	10.4599ms	180.2444 μ s	278.9824ms	4.5712ms	51.4984 μ s
	2116	28.7082ms	370.9793 μ s	642.7033ms	7.7991ms	50.7832 μ s
	5184	142.9813ms	1.1504ms	2.4227s	4.1821ms	45.0611 μ s
	10201	447.1834ms	3.0591ms	5.8252s	10.7284ms	67.7109 μ s
	20164	1.9033s	6.9902ms	15.9052s	13.3739ms	103.7121 μ s
HODLR	1089	14.4815ms	232.6965 μ s	287.4746ms	4.0405ms	84.1618 μ s
	2116	22.3615ms	375.9861 μ s	698.2346ms	4.3597ms	76.5324 μ s
	5184	151.0005ms	1.7681ms	3.0918s	3.9043ms	66.5188 μ s
	10201	577.9347ms	4.2386ms	8.3628s	11.8380ms	147.3427 μ s
	20164	2.7849s	11.8909ms	26.7882s	8.3604ms	129.2229 μ s

Exponential model

Table 7: 2D exponential corr=0.1 — hodlr-branch build

method	n	1st	2nd	KL	fit	regr
LAPACK	1089	93.1802ms	301.3611 μ s	3.0905s	3.9027ms	356.9126 μ s
	2116	35.3630ms	391.0065 μ s	—	3.8285ms	237.7033 μ s
	5184	—	—	—	2.8937ms	239.1338 μ s
	10201	—	—	—	2.5241ms	180.0060 μ s
	20164	—	—	—	6.1731ms	307.0831 μ s
HMAT	1089	21.4729ms	331.6402 μ s	119.3457ms	5.0552ms	58.1741 μ s
	2116	85.8889ms	780.8208 μ s	208.1223ms	3.9816ms	52.2137 μ s
	5184	248.5654ms	1.2915ms	458.4751ms	3.9060ms	62.2272 μ s
	10201	472.9950ms	3.1650ms	944.3631ms	5.1637ms	59.8431 μ s
	20164	1.3750s	6.7768ms	2.3693s	6.8345ms	77.9629 μ s

continued

method	n	1st	2nd	KL	fit	regr
HMAT_HODLR	1089	16.1004ms	244.6175 μ s	91.3460ms	4.2942ms	53.6442 μ s
	2116	47.7505ms	453.4721 μ s	162.3869ms	3.6693ms	54.3594 μ s
	5184	215.7633ms	1.2181ms	457.3236ms	3.2597ms	40.5312 μ s
	10201	533.9353ms	3.1064ms	964.3471ms	5.1498ms	63.8962 μ s
	20164	1.3450s	7.0181ms	2.2762s	6.7022ms	73.1945 μ s
HODLR	1089	18.6667ms	253.4389 μ s	97.3985ms	6.8409ms	121.1166 μ s
	2116	59.7475ms	561.7142 μ s	221.4882ms	6.4325ms	89.4070 μ s
	5184	297.9746ms	2.0351ms	768.9490ms	6.1667ms	90.8375 μ s
	10201	1.1620s	4.9434ms	2.3160s	15.6081ms	180.9597 μ s
	20164	3.9996s	13.1323ms	6.8274s	21.3306ms	254.3926 μ s

Table 8: 2D exponential corr=0.01 — hodlr-branch build

method	n	1st	2nd	KL	fit	regr
LAPACK	1089	23.2470ms	5.9452ms	3.6674s	4.6508ms	391.4833 μ s
	2116	35.7690ms	387.1918 μ s	—	3.4153ms	254.1542 μ s
	5184	—	—	—	2.7797ms	217.1993 μ s
	10201	—	—	—	2.4688ms	217.9146 μ s
	20164	—	—	—	4.7383ms	278.9497 μ s
HMAT	1089	13.6392ms	319.4809 μ s	342.9539ms	2.9492ms	49.8295 μ s
	2116	43.8447ms	612.9742 μ s	710.4399ms	3.0992ms	48.3990 μ s
	5184	125.5546ms	1.2679ms	1.8568s	2.6391ms	49.3526 μ s
	10201	423.6612ms	3.1271ms	5.0708s	3.9179ms	58.4126 μ s
	20164	1.6045s	6.9604ms	11.2271s	5.9586ms	74.6250 μ s
HMAT_HODLR	1089	8.2984ms	198.8411 μ s	263.4544ms	2.8672ms	66.2804 μ s
	2116	22.9244ms	314.9509 μ s	633.9145ms	3.0372ms	58.4126 μ s
	5184	108.2597ms	1.0395ms	1.8303s	2.6355ms	41.0080 μ s
	10201	501.6556ms	3.2732ms	5.1099s	4.8151ms	87.9765 μ s
	20164	1.6123s	6.7897ms	11.1985s	5.7871ms	81.5392 μ s
HODLR	1089	9.5596ms	157.8331 μ s	279.1872ms	5.3997ms	78.9165 μ s
	2116	32.1107ms	457.2868 μ s	715.2326ms	4.4754ms	77.7245 μ s
	5184	142.8716ms	1.6279ms	2.4711s	4.5161ms	86.0691 μ s
	10201	632.7877ms	4.2012ms	7.5054s	8.6846ms	189.3044 μ s
	20164	2.2872s	10.9966ms	18.2578s	11.8845ms	138.9980 μ s

3.3 3D

Matern model

Table 9: 3D matern corr=0.1 — hodlr-branch build

method	n	1st	2nd	KL	fit	regr
LAPACK	1331	53.8659ms	360.4889 μ s	9.0479s	5.5592ms	455.8563 μ s
	2744	462.7042ms	826.5972 μ s	—	6.2332ms	407.6958 μ s

continued

method	n	1st	2nd	KL	fit	regr
HMAT	5832	—	—	—	2.9371ms	280.1418 μ s
	12167	—	—	—	2.6259ms	303.5069 μ s
	21952	—	—	—	5.7557ms	370.5025 μ s
	1331	91.3033ms	558.6147 μ s	270.7613ms	9.4831ms	105.8578 μ s
	2744	407.0199ms	856.6380 μ s	604.4471ms	7.7419ms	59.1278 μ s
HMAT_HODLR	5832	1.6701s	2.6238ms	2.0697s	7.4556ms	45.7764 μ s
	12167	4.1772s	7.3397ms	4.1881s	9.5580ms	110.1494 μ s
	21952	37.5214s	18.6536ms	28.6521s	13.2298ms	92.7448 μ s
	1331	40.2195ms	257.0152 μ s	231.7562ms	8.9335ms	91.3143 μ s
	2744	323.6511ms	857.3532 μ s	629.5331ms	7.9749ms	66.7572 μ s
HODLR	5832	1.8795s	2.7192ms	2.0703s	7.3562ms	47.9221 μ s
	12167	4.2626s	7.3035ms	4.2227s	9.5074ms	109.1957 μ s
	21952	38.0825s	18.8360ms	34.2415s	13.5102ms	103.2352 μ s
	1331	57.0161ms	418.4246 μ s	296.5469ms	12.6035ms	188.3507 μ s
	2744	266.3779ms	1.3635ms	815.4907ms	8.3797ms	121.3551 μ s
	5832	2.0469s	5.0681ms	4.1053s	8.0109ms	119.4477 μ s
	12167	14.7858s	18.2786ms	21.9035s	36.2575ms	245.3327 μ s
	21952	53.3999s	45.7239ms	1.2m	24.4787ms	252.9621 μ s

Table 10: 3D matern corr=0.01 — hodlr-branch build

method	n	1st	2nd	KL	fit	regr
LAPACK	1331	467.5014ms	364.7804 μ s	5.9962s	63.4389ms	475.8835 μ s
	2744	2.0587s	783.6819 μ s	—	55.9225ms	433.4450 μ s
	5832	—	—	—	8.5311ms	359.7736 μ s
	12167	—	—	—	5.0337ms	171.6614 μ s
	21952	—	—	—	8.8177ms	279.4266 μ s
HMAT	1331	156.5142ms	410.7952 μ s	346.4780ms	42.2583ms	103.7121 μ s
	2744	329.7405ms	848.0549 μ s	965.5001ms	4.3666ms	44.1074 μ s
	5832	795.3138ms	2.3932ms	3.0174s	5.7445ms	82.4928 μ s
	12167	1.2562s	5.2769ms	7.6111s	31.7142ms	133.9912 μ s
	21952	22.1231s	14.5910ms	30.1350s	14.0910ms	109.4341 μ s
HMAT_HODLR	1331	107.6074ms	277.2808 μ s	337.3742ms	42.0544ms	101.8047 μ s
	2744	233.2449ms	741.4818 μ s	937.7890ms	8.3117ms	54.1210 μ s
	5832	1.0466s	2.3155ms	3.1046s	5.8115ms	45.7764 μ s
	12167	1.3320s	5.6498ms	7.4636s	30.7505ms	124.2161 μ s
	21952	19.8267s	13.5181ms	33.2641s	13.4277ms	112.5336 μ s
HODLR	1331	11.4372ms	213.1462 μ s	306.6149ms	9.6381ms	113.0104 μ s
	2744	18.3754ms	432.2529 μ s	804.8606ms	6.5238ms	88.2149 μ s
	5832	42.5775ms	1.0483ms	1.9486s	6.4077ms	63.1809 μ s
	12167	561.5752ms	4.4830ms	7.3531s	47.7893ms	209.0931 μ s
	21952	9.7164s	18.3160ms	37.1015s	14.7843ms	186.2049 μ s

Exponential model

Table 11: 3D exponential corr=0.1 — hodlr-branch build

method	n	1st	2nd	KL	fit	regr
LAPACK	1331	51.6911ms	321.3882 μ s	5.0258s	4.9047ms	406.7421 μ s
	2744	123.6432ms	680.9235 μ s	—	5.6293ms	347.3759 μ s
	5832	—	—	—	2.8634ms	238.1802 μ s
	12167	—	—	—	2.4822ms	179.2908 μ s
	21952	—	—	—	5.9321ms	304.9374 μ s
HMAT	1331	61.0328ms	434.8755 μ s	235.0338ms	7.2508ms	104.1889 μ s
	2744	479.5606ms	900.5070 μ s	488.3468ms	10.1733ms	56.7436 μ s
	5832	1.7959s	2.6908ms	1.8576s	8.0841ms	65.5651 μ s
	12167	3.5599s	6.8259ms	3.3941s	8.5454ms	104.1889 μ s
	21952	33.5580s	17.6041ms	23.8601s	15.1646ms	121.1166 μ s
HMAT_HODLR	1331	42.9518ms	328.5408 μ s	189.8410ms	6.6652ms	96.3211 μ s
	2744	352.8013ms	786.5429 μ s	523.8967ms	10.0677ms	64.6114 μ s
	5832	1.6672s	2.6016ms	1.8622s	8.7674ms	53.4058 μ s
	12167	3.6305s	7.4861ms	3.3650s	8.4565ms	87.4996 μ s
	21952	30.4314s	17.2007ms	25.3626s	15.5098ms	108.9573 μ s
HODLR	1331	68.7754ms	390.2912 μ s	267.8723ms	13.1667ms	185.2512 μ s
	2744	344.9678ms	1.5299ms	838.9211ms	10.8173ms	144.9585 μ s
	5832	2.2346s	5.4088ms	3.7833s	10.1397ms	112.7720 μ s
	12167	13.4832s	16.3355ms	18.8895s	18.4433ms	225.7824 μ s
	21952	54.7557s	44.7559ms	1.1m	21.3869ms	305.6526 μ s

Table 12: 3D exponential corr=0.01 — hodlr-branch build

method	n	1st	2nd	KL	fit	regr
LAPACK	1331	37.3495ms	7.4260ms	6.7734s	4.8294ms	452.7569 μ s
	2744	115.5274ms	656.1279 μ s	—	5.1622ms	334.5013 μ s
	5832	—	—	—	2.6560ms	182.8671 μ s
	12167	—	—	—	2.5995ms	166.1777 μ s
	21952	—	—	—	6.3515ms	330.6866 μ s
HMAT	1331	32.6090ms	392.1986 μ s	350.9467ms	7.0925ms	152.1111 μ s
	2744	325.5160ms	715.4942 μ s	809.9883ms	3.1631ms	39.5775 μ s
	5832	1.0218s	2.5244ms	2.7998s	2.5828ms	41.2464 μ s
	12167	1.2890s	5.5177ms	6.9915s	7.3378ms	102.0432 μ s
	21952	16.3447s	12.9969ms	31.3547s	7.6063ms	106.8115 μ s
HMAT_HODLR	1331	23.0911ms	243.9022 μ s	297.4274ms	6.5870ms	84.8770 μ s
	2744	207.8104ms	855.2074 μ s	834.6782ms	2.9163ms	43.3922 μ s
	5832	993.2799ms	2.4061ms	2.8279s	2.5418ms	49.5911 μ s
	12167	1.4402s	5.6927ms	7.2602s	7.5951ms	115.6330 μ s
	21952	18.4766s	13.6116ms	29.8826s	7.9799ms	114.2025 μ s
HODLR	1331	10.0069ms	205.9937 μ s	266.4101ms	5.6715ms	132.0839 μ s
	2744	85.4776ms	853.3001 μ s	828.5685ms	4.5142ms	85.8307 μ s
	5832	394.8166ms	2.7306ms	3.0516s	4.6666ms	92.0296 μ s
	12167	828.2123ms	5.4245ms	8.6225s	14.7405ms	178.0987 μ s
	21952	5.2138s	16.3009ms	27.9567s	17.8001ms	246.7632 μ s

3.4 4D

Matern model

Table 13: 4D matern corr=0.1 — hodlr-branch build

method	n	1st	2nd	KL	fit	regr
LAPACK	2401	288.2361ms	682.1156 μ s	—	10.8445ms	472.5456 μ s
	4096	420.8722ms	1.6065ms	—	3.5322ms	294.9238 μ s
	6561	—	—	—	2.4488ms	155.6873 μ s
	14641	—	—	—	2.2948ms	167.3698 μ s
	28561	—	—	—	3.1760ms	226.7361 μ s
HMAT	2401	313.8375ms	900.2686 μ s	987.3374ms	7.2718ms	97.2748 μ s
	4096	2.3271s	1.7982ms	2.8507s	8.4047ms	39.1006 μ s
	6561	2.1178s	4.5502ms	3.9228s	4.7688ms	42.9153 μ s
	14641	14.0385s	14.4463ms	12.6740s	13.4435ms	118.7325 μ s
	28561	1.2m	41.8508ms	39.1950s	9.8696ms	110.8646 μ s
HMAT_HODLR	2401	210.0275ms	1.0121ms	1.0148s	7.4635ms	86.0691 μ s
	4096	2.2481s	1.8084ms	2.7569s	8.8439ms	44.1074 μ s
	6561	2.0056s	4.6463ms	3.9267s	4.9455ms	46.9685 μ s
	14641	14.3498s	14.5252ms	12.7571s	14.3251ms	112.0567 μ s
	28561	1.1m	42.7136ms	36.8834s	9.9742ms	113.7257 μ s
HODLR	2401	253.2983ms	1.2739ms	1.3935s	9.9735ms	163.0783 μ s
	4096	1.7677s	4.4272ms	4.7282s	8.3697ms	99.8974 μ s
	6561	6.0437s	8.3904ms	11.6255s	7.5634ms	97.7516 μ s
	14641	1.0m	38.4958ms	1.4m	27.5059ms	345.2301 μ s
	28561	5.5m	124.2776ms	6.5m	24.3654ms	240.3259 μ s

Table 14: 4D matern corr=0.01 — hodlr-branch build

method	n	1st	2nd	KL	fit	regr
LAPACK	2401	3.2781s	1.8308ms	—	142.9334ms	708.3416 μ s
	4096	11.3590s	6.3510ms	—	41.1587ms	494.7186 μ s
	6561	—	—	—	7.2768ms	468.7309 μ s
	14641	—	—	—	6.9668ms	183.3439 μ s
	28561	—	—	—	59.5534ms	334.5013 μ s
HMAT	2401	930.3677ms	999.9275 μ s	1.2575s	51.3246ms	107.2884 μ s
	4096	2.1514s	1.7402ms	3.0441s	6.1345ms	65.0883 μ s
	6561	—	—	—	31.0545ms	60.3199 μ s
	14641	4.6859s	10.1156ms	14.9175s	25.8245ms	109.1957 μ s
	28561	15.6725s	26.4904ms	47.9960s	58.3122ms	133.5144 μ s
HMAT_HODLR	2401	876.5028ms	897.8844 μ s	1.2887s	52.7368ms	99.4205 μ s
	4096	2.3898s	1.7650ms	3.0059s	6.0623ms	46.7300 μ s
	6561	—	—	—	31.8134ms	72.0024 μ s
	14641	4.4546s	10.2549ms	15.0471s	26.9020ms	125.8850 μ s
	28561	15.0962s	25.3327ms	48.9037s	56.1926ms	142.5743 μ s
HODLR	2401	21.9500ms	386.9534 μ s	1.1188s	13.0639ms	128.0308 μ s
	4096	25.9686ms	630.1403 μ s	2.2836s	14.6258ms	87.4996 μ s
	6561	56.9479ms	1.3635ms	4.1159s	17.0429ms	86.0691 μ s
	14641	191.5581ms	3.4595ms	13.1743s	14.4191ms	187.1586 μ s

continued

method	n	1st	2nd	KL	fit	regr
	28561	289.9911ms	6.2542ms	29.7540s	18.8386ms	189.0659 μ s

Exponential model

Table 15: 4D exponential corr=0.1 — hodlr-branch build

method	n	1st	2nd	KL	fit	regr
LAPACK	2401	402.0414ms	3.8712ms	—	8.8272ms	639.2002 μ s
	4096	619.5028ms	2.0621ms	—	7.6385ms	288.0096 μ s
	6561	—	—	—	2.4610ms	167.3698 μ s
	14641	—	—	—	2.4724ms	182.6286 μ s
	28561	—	—	—	3.5710ms	222.6830 μ s
HMAT	2401	485.1787ms	1.0889ms	909.4236ms	5.7256ms	87.9765 μ s
	4096	2.6624s	1.9388ms	2.6400s	9.6068ms	41.4848 μ s
	6561	3.5725s	5.3520ms	3.9592s	3.8915ms	59.3662 μ s
	14641	22.6552s	16.5522ms	13.8435s	13.2675ms	90.3606 μ s
	28561	1.4m	44.9786ms	38.9344s	9.3765ms	141.3822 μ s
HMAT_HODLR	2401	369.4057ms	1.0355ms	911.9034ms	5.6779ms	82.7312 μ s
	4096	2.6152s	1.9414ms	2.5648s	10.1202ms	41.4848 μ s
	6561	3.5938s	5.2688ms	3.8680s	3.7231ms	49.8295 μ s
	14641	22.2947s	16.9730ms	13.8339s	18.3058ms	110.8646 μ s
	28561	1.4m	44.3966ms	38.4688s	8.3506ms	110.6262 μ s
HODLR	2401	456.6236ms	1.5626ms	1.5914s	15.8746ms	210.0468 μ s
	4096	1.9581s	4.5688ms	4.5412s	10.8716ms	116.1098 μ s
	6561	7.3406s	9.5956ms	12.4509s	12.0142ms	146.1506 μ s
	14641	1.2m	39.4874ms	1.5m	34.3482ms	318.7656 μ s
	28561	6.6m	135.4160ms	7.5m	29.3164ms	309.9442 μ s

Table 16: 4D exponential corr=0.01 — hodlr-branch build

method	n	1st	2nd	KL	fit	regr
LAPACK	2401	350.4376ms	1.8454ms	—	8.3802ms	526.1898 μ s
	4096	262.3246ms	2.2638ms	—	5.9886ms	339.0312 μ s
	6561	—	—	—	2.4934ms	185.4897 μ s
	14641	—	—	—	2.6395ms	174.5224 μ s
	28561	—	—	—	3.2237ms	240.8028 μ s
HMAT	2401	198.1721ms	891.4471 μ s	1.2989s	5.7049ms	86.7844 μ s
	4096	2.1014s	1.6718ms	3.6609s	8.1570ms	49.5911 μ s
	6561	872.7279ms	3.9206ms	5.2524s	3.8064ms	46.7300 μ s
	14641	4.6492s	10.5054ms	16.7476s	8.7130ms	117.5404 μ s
	28561	20.1209s	27.1673ms	43.4052s	8.3973ms	109.9110 μ s
HMAT_HODLR	2401	123.5490ms	841.8560 μ s	1.2734s	5.8177ms	108.7189 μ s
	4096	2.3658s	1.7681ms	3.4508s	8.3935ms	43.8690 μ s
	6561	841.8539ms	4.3154ms	5.1039s	3.7208ms	61.5120 μ s

continued

method	n	1st	2nd	KL	fit	regr
HODLR	14641	4.7076s	10.8385ms	16.5923s	9.9828ms	96.5595 μ s
	28561	19.9615s	28.2931ms	44.3659s	8.9035ms	133.5144 μ s
	2401	17.8931ms	377.4166 μ s	1.0589s	5.7857ms	109.9110 μ s
	4096	26.6359ms	623.2262 μ s	2.2383s	4.4353ms	94.4138 μ s
	6561	60.5278ms	1.2770ms	3.8200s	5.3985ms	87.9765 μ s
	14641	209.6107ms	3.4010ms	11.2221s	12.6488ms	178.3371 μ s
	28561	321.8179ms	6.5174ms	25.8582s	15.6469ms	178.5755 μ s

4 Direct comparison: self-contained HODLR vs hmat-oss HODLR (PR #3272)

Highlight of this report. Both HODLR variants were measured in the same benchmark session (2026-09-13) on the same machine with identical sweeps:

- HODLR (self-contained, this branch) — `hodlr-branch` install;
- HMAT_HODLR (hmat-oss HODLR, PR #3272) — `pr3272` install. On the branch build of 09-13 it used the HODLR admissibility rule with the LLt factorization (the dedicated ‘hodlr’ factorization enum was only accepted by the PR build).

The tables below give per-configuration first-realization and KL timings with the ratio HODLR/HMAT_HODLR (< 1 : self-contained faster, > 1 : hmat-oss faster).

Table 17: 1D matern corr=0.1: HODLR (branch) vs HMAT_HODLR (PR#3272)

n	1st br	1st pr	ratio	KL br	KL pr	ratio
1001	10.5488ms	11.6405ms	0.91 $\times$	26.6383ms	33.4082ms	0.80 $\times$
2001	11.9703ms	20.9384ms	0.57 $\times$	52.2609ms	55.3608ms	0.94 $\times$
5001	44.0936ms	36.6611ms	1.20 $\times$	134.0339ms	118.6428ms	1.13 $\times$
10001	282.3546ms	137.4211ms	2.05 $\times$	3.5674s	289.8824ms	12.31 $\times$
20001	26.0115s	256.9916ms	101.22 $\times$	26.3912s	599.9017ms	43.99 $\times$

Table 18: 1D matern corr=0.01: HODLR (branch) vs HMAT_HODLR (PR#3272)

n	1st br	1st pr	ratio	KL br	KL pr	ratio
1001	6.4704ms	8.9042ms	0.73 $\times$	147.0857ms	143.9044ms	1.02 $\times$
2001	12.4528ms	17.7395ms	0.70 $\times$	290.0660ms	259.9883ms	1.12 $\times$
5001	30.5357ms	33.0706ms	0.92 $\times$	797.7509ms	656.7769ms	1.21 $\times$
10001	113.8282ms	137.3682ms	0.83 $\times$	1.5992s	1.3051s	1.23 $\times$
20001	195.0102ms	246.9075ms	0.79 $\times$	3.3123s	2.7327s	1.21 $\times$

Table 19: 1D exponential corr=0.1: HODLR (branch) vs HMAT_HODLR (PR#3272)

n	1st br	1st pr	ratio	KL br	KL pr	ratio
1001	9.0213ms	6.0246ms	1.50 $\times$	43.6378ms	44.2808ms	0.99 $\times$
2001	13.4776ms	9.8855ms	1.36 $\times$	71.5711ms	67.2607ms	1.06 $\times$
5001	29.0082ms	16.3558ms	1.77 $\times$	176.3484ms	139.1108ms	1.27 $\times$
10001	111.4185ms	58.9273ms	1.89 $\times$	421.5465ms	408.1161ms	1.03 $\times$
20001	209.5237ms	127.4259ms	1.64 $\times$	836.4038ms	662.4277ms	1.26 $\times$

Table 20: 1D exponential corr=0.01: HODLR (branch) vs HMAT_HODLR (PR#3272)

n	1st br	1st pr	ratio	KL br	KL pr	ratio
1001	7.0734ms	4.4770ms	1.58 $\times$	116.0200ms	113.9324ms	1.02 $\times$
2001	12.8524ms	8.5020ms	1.51 $\times$	216.7389ms	193.5139ms	1.12 $\times$
5001	31.1267ms	15.3091ms	2.03 $\times$	584.2237ms	490.9940ms	1.19 $\times$
10001	111.7826ms	67.4462ms	1.66 $\times$	1.1775s	1.0623s	1.11 $\times$
20001	210.1967ms	131.1626ms	1.60 $\times$	2.4704s	1.9613s	1.26 $\times$

Table 21: 2D matern corr=0.1: HODLR (branch) vs HMAT_HODLR (PR#3272)

n	1st br	1st pr	ratio	KL br	KL pr	ratio
1089	18.3501ms	32.8791ms	0.56 $\times$	112.0167ms	100.6176ms	1.11 $\times$
2116	55.7771ms	78.1102ms	0.71 $\times$	232.1019ms	189.4584ms	1.23 $\times$
5184	273.1204ms	247.9246ms	1.10 $\times$	843.8439ms	502.6531ms	1.68 $\times$
10201	905.1511ms	636.5881ms	1.42 $\times$	2.1841s	1.1505s	1.90 $\times$
20164	2.6493s	1.1509s	2.30 $\times$	5.5858s	2.4208s	2.31 $\times$

Table 22: 2D matern corr=0.01: HODLR (branch) vs HMAT_HODLR (PR#3272)

n	1st br	1st pr	ratio	KL br	KL pr	ratio
1089	14.4815ms	9.9273ms	1.46 $\times$	287.4746ms	249.1457ms	1.15 $\times$
2116	22.3615ms	32.0807ms	0.70 $\times$	698.2346ms	630.5602ms	1.11 $\times$
5184	151.0005ms	138.0615ms	1.09 $\times$	3.0918s	2.2512s	1.37 $\times$
10201	577.9347ms	565.6343ms	1.02 $\times$	8.3628s	5.3782s	1.55 $\times$
20164	2.7849s	3.4210s	0.81 $\times$	26.7882s	17.7906s	1.51 $\times$

Table 23: 2D exponential corr=0.1: HODLR (branch) vs HMAT_HODLR (PR#3272)

n	1st br	1st pr	ratio	KL br	KL pr	ratio
1089	18.6667ms	22.5365ms	0.83 $\times$	97.3985ms	85.8605ms	1.13 $\times$
2116	59.7475ms	53.0508ms	1.13 $\times$	221.4882ms	147.4812ms	1.50 $\times$
5184	297.9746ms	209.7712ms	1.42 $\times$	768.9490ms	432.0564ms	1.78 $\times$
10201	1.1620s	696.1195ms	1.67 $\times$	2.3160s	1.0524s	2.20 $\times$
20164	3.9996s	1.8531s	2.16 $\times$	6.8274s	2.6157s	2.61 $\times$

Table 24: 2D exponential corr=0.01: HODLR (branch) vs HMAT_HODLR (PR#3272)

n	1st br	1st pr	ratio	KL br	KL pr	ratio
1089	9.5596ms	8.2629ms	1.16 $\times$	279.1872ms	224.7763ms	1.24 $\times$
2116	32.1107ms	23.5248ms	1.36 $\times$	715.2326ms	575.5746ms	1.24 $\times$
5184	142.8716ms	109.0174ms	1.31 $\times$	2.4711s	1.7480s	1.41 $\times$
10201	632.7877ms	665.7209ms	0.95 $\times$	7.5054s	4.7596s	1.58 $\times$
20164	2.2872s	2.4964s	0.92 $\times$	18.2578s	11.7793s	1.55 $\times$

Table 25: 3D matern corr=0.1: HODLR (branch) vs HMAT_HODLR (PR#3272)

n	1st br	1st pr	ratio	KL br	KL pr	ratio
1331	57.0161ms	65.9263ms	0.86 $\times$	296.5469ms	221.2462ms	1.34 $\times$
2744	266.3779ms	384.3560ms	0.69 $\times$	815.4907ms	615.7634ms	1.32 $\times$
5832	2.0469s	2.3730s	0.86 $\times$	4.1053s	2.4816s	1.65 $\times$
12167	14.7858s	10.2520s	1.44 $\times$	21.9035s	8.8539s	2.47 $\times$
21952	53.3999s	40.3381s	1.32 $\times$	1.2m	31.6960s	2.21 $\times$

Table 26: 3D matern corr=0.01: HODLR (branch) vs HMAT_HODLR (PR#3272)

n	1st br	1st pr	ratio	KL br	KL pr	ratio
1331	11.4372ms	48.0986ms	0.24 $\times$	306.6149ms	322.1159ms	0.95 $\times$
2744	18.3754ms	239.2638ms	0.08 $\times$	804.8606ms	881.0570ms	0.91 $\times$
5832	42.5775ms	791.8546ms	0.05 $\times$	1.9486s	2.6092s	0.75 $\times$
12167	561.5752ms	3.3441s	0.17 $\times$	7.3531s	8.8620s	0.83 $\times$
21952	9.7164s	16.6647s	0.58 $\times$	37.1015s	33.0586s	1.12 $\times$

Table 27: 3D exponential corr=0.1: HODLR (branch) vs HMAT_HODLR (PR#3272)

n	1st br	1st pr	ratio	KL br	KL pr	ratio
1331	68.7754ms	97.4934ms	0.71 $\times$	267.8723ms	208.1840ms	1.29 $\times$
2744	344.9678ms	403.7175ms	0.85 $\times$	838.9211ms	550.9338ms	1.52 $\times$
5832	2.2346s	1.8365s	1.22 $\times$	3.7833s	1.9238s	1.97 $\times$
12167	13.4832s	9.2150s	1.46 $\times$	18.8895s	8.1155s	2.33 $\times$
21952	54.7557s	31.3624s	1.75 $\times$	1.1m	26.2262s	2.58 $\times$

Table 28: 3D exponential corr=0.01: HODLR (branch) vs HMAT_HODLR (PR#3272)

n	1st br	1st pr	ratio	KL br	KL pr	ratio
1331	10.0069ms	44.1008ms	0.23 $\times$	266.4101ms	276.0582ms	0.97 $\times$
2744	85.4776ms	234.9958ms	0.36 $\times$	828.5685ms	741.0998ms	1.12 $\times$
5832	394.8166ms	1.0668s	0.37 $\times$	3.0516s	2.8227s	1.08 $\times$
12167	828.2123ms	4.7037s	0.18 $\times$	8.6225s	7.6223s	1.13 $\times$
21952	5.2138s	22.0376s	0.24 $\times$	27.9567s	25.6575s	1.09 $\times$

Table 29: 4D matern corr=0.1: HODLR (branch) vs HMAT_HODLR (PR#3272)

n	1st br	1st pr	ratio	KL br	KL pr	ratio
2401	253.2983ms	710.4745ms	0.36 ×	1.3935s	1.2030s	1.16 ×
4096	1.7677s	2.4579s	0.72 ×	4.7282s	2.8410s	1.66 ×
6561	6.0437s	6.6821s	0.90 ×	11.6255s	7.1035s	1.64 ×
14641	1.0m	47.0867s	1.29 ×	1.4m	32.6476s	2.49 ×
28561	5.5m	4.2m	1.31 ×	6.5m	2.8m	2.35 ×

Table 30: 4D matern corr=0.01: HODLR (branch) vs HMAT_HODLR (PR#3272)

n	1st br	1st pr	ratio	KL br	KL pr	ratio
2401	21.9500ms	460.6719ms	0.05 ×	1.1188s	1.3613s	0.82 ×
4096	25.9686ms	2.0907s	0.01 ×	2.2836s	3.6726s	0.62 ×
6561	56.9479ms	6.3089s	0.01 ×	4.1159s	9.1570s	0.45 ×
14641	191.5581ms	28.9026s	0.01 ×	13.1743s	28.4962s	0.46 ×
28561	289.9911ms	2.5m	0.00 ×	29.7540s	2.4m	0.21 ×

Table 31: 4D exponential corr=0.1: HODLR (branch) vs HMAT_HODLR (PR#3272)

n	1st br	1st pr	ratio	KL br	KL pr	ratio
2401	456.6236ms	780.2052ms	0.59 ×	1.5914s	1.0519s	1.51 ×
4096	1.9581s	2.5286s	0.77 ×	4.5412s	2.5871s	1.76 ×
6561	7.3406s	8.8443s	0.83 ×	12.4509s	6.8654s	1.81 ×
14641	1.2m	1.1m	1.01 ×	1.5m	43.1228s	2.02 ×
28561	6.6m	5.9m	1.11 ×	7.5m	4.0m	1.86 ×

Table 32: 4D exponential corr=0.01: HODLR (branch) vs HMAT_HODLR (PR#3272)

n	1st br	1st pr	ratio	KL br	KL pr	ratio
2401	17.8931ms	483.7897ms	0.04 ×	1.0589s	1.3605s	0.78 ×
4096	26.6359ms	1.9174s	0.01 ×	2.2383s	3.4550s	0.65 ×
6561	60.5278ms	5.0895s	0.01 ×	3.8200s	7.8824s	0.48 ×
14641	209.6107ms	37.6573s	0.01 ×	11.2221s	40.2980s	0.28 ×
28561	321.8179ms	2.7m	0.00 ×	25.8582s	3.0m	0.14 ×

4.1 Reading the comparison

Large correlation length (corr=0.1), 1D

The self-contained HODLR collapses at $n \geq 10000$: $n = 20001$ first realization 26.0s and KL 26.4s vs 0.26s and 0.60s for HMAT_HODLR — a $44\text{--}101\times$ slowdown. This is the pre-fix corrupt-heal regime (Section 2); after the fix the branch reaches ≈ 1.8 s factorize at $n = 20002$ (idle gate), which restores parity with the hmat-oss backend at this size.

Short correlation length (corr=0.01), high dimensions

The picture inverts: the self-contained implementation is *much faster*. At 4D corr=0.01 $n = 28561$: first realization 0.29s vs 152.3s ($\approx 490\times$ faster) and KL 29.8s vs 144.4s ($\approx 4.9\times$). At 3D/2D corr=0.01 the branch is $1.2\text{--}6\times$ faster. hmat-oss HODLR pays heavily for low-correlation data in high dimensions (large ranks survive its recompression), whereas the self-contained low-rank recompression keeps ranks tight.

KL algorithm

For corr=0.1 the hmat-oss HODLR backend is generally $1.1\text{--}2.6\times$ faster on KL; for corr=0.01 in high dimensions the self-contained backend leads (ratios down to $0.14\times$, i.e. $7\times$ faster).

5 Overall conclusions

1. **Accuracy:** the one measured accuracy gate passes ($\text{relerr} = 4.7 \times 10^{-3} < 10^{-2}$, $n = 4001$); the pre-fix corrupt regime is eliminated by the pivot-floor heal. An accuracy figure for the hmat-oss HODLR backend is *missing* — only speed was compared.
2. **Speed:** self-contained HODLR is competitive with or faster than HMAT/HMAT_HODLR up to $n \sim 10^4$, and dramatically faster than hmat-oss HODLR for short correlation lengths in high dimensions; its few large- n /long-corr slow entries are the (fixed) corrupt-heal artefact and must be re-measured after the 2026-09-16 fix.
3. **Metrics to complete:** re-run `benchmark_gp_full.py --accuracy` and persist the Litvinenko suite (`rel_norm2`, `gap2`, `KL`, `logdet`); capture a dense-vs-HMAT_HODLR accuracy point to complete the implementation-by-implementation accuracy comparison.